MediTrack

Software Requirements Specification

A Secure E-Commerce Portal for a Healthcare Store

Prepared per IEEE 830 SRS structure

Version	1.0
Modules Covered	Store, Cart/Wishlist, Billing, Customer/CRM, Security
Primary Stack	React, Node.js/Express, MongoDB

This document will be revised as requirements are refined during development and after the predecessor group's security audit.

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1 Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the functional and non-functional requirements for MediTrack, a secure e-commerce portal for a pharmacy store.

The document is intended for the development team, project evaluators, and the predecessor group responsible for the security audit.

The requirements described in this document are intended to serve as testable contracts for the implementation and evaluation of the system.

1.2 Scope

MediTrack is designed as a secure pharmacy e-commerce platform providing the following capabilities:

- A medicine catalog containing approximately 50 items with live stock visibility.
- A shopping Cart with short-lived stock holds and a separate Wishlist.
- Prescription-gated checkout for medicines requiring a prescription.
- Billing through a sandbox payment gateway.
- Customer accounts, purchase history, MediPoints, and refill reminders.
- Role-Based Access Control (RBAC) and append-only audit logging.
- A tracer-bullet development philosophy in which a thin end-to-end workflow is implemented first: browse → hold stock → verify prescription → pay → confirm.

1.3 Definitions and Acronyms

Term	Definition
SRS	Software Requirements Specification.
RBAC	Role-Based Access Control used to restrict system operations according to user roles.
JWT	JSON Web Token used for authenticated sessions.
TTL	Time To Live; the period for which a temporary stock reservation remains valid.
CRM	Customer Relationship Management.
SCM	Supply Chain Management.
Rx	Prescription issued by an authorized medical practitioner.
Soft Hold	A temporary reservation of inventory associated with a customer's cart.
MediPoints	Reward points credited to a customer after successful payment.

1.4 References

- IEEE Std 830-1998, IEEE Recommended Practice for Software Requirements Specifications.
- MediTrack Architecture & Planning Document.
- High-Level Design UML Component Diagram.

2 Overall Description

2.1 Product Perspective

MediTrack is designed as a standalone pharmacy e-commerce system following a three-tier architecture.

The major architectural components are:

- React-based client application.
- Node.js/Express backend API.
- MongoDB database.
- Third-party sandbox payment gateway.

The system separates the user interface, application logic, and persistent data layers.

2.2 User Classes

The system supports the following major user classes.

User Class	Description
Customer	A user who browses medicines, manages a cart and wishlist, uploads prescriptions where required, places orders, makes payments, and views purchase history.
Inventory Manager	A privileged user responsible for managing medicines, inventory, prescription verification, and inventory-related operations.
Auditor	A user associated with the predecessor group who reviews security controls and audit records.

2.3 Operating Environment

The system is intended to operate in the following environment:

- Modern web browsers.
- HTTPS-enabled communication.
- Node.js server environment.
- MongoDB database environment.
- Sandbox payment gateway.

2.4 Design Constraints

The project is developed under the following constraints:

- The implementation is limited to one academic semester.
- The development team consists of three members.

- Payment processing is restricted to sandbox/test mode.
- Hardware-dependent QR/RFID functionality is simulated.

2.5 Assumptions

The following assumptions are used during system design:

- India does not, by default, permit standing refill authorizations for the intended workflow.
- A stable Internet connection is available during checkout.
- Doctors are available for teleconsultation when required.

3 Specific Requirements

3.1 Functional Requirements

3.1.1 Store and Cart Requirements

ID	Priority	Requirement
FR-1.1	High	The system shall replenish medicine inventory when the stock level reaches the configured watermark of 10 units, restoring the stock level to 100 units.
FR-1.2	High	The system shall create a short-lived soft hold when an item is added to the cart. The stock check and reservation shall be performed atomically to prevent overselling.
FR-1.3	High	The system shall automatically release an expired soft hold and make the corresponding inventory available again.
FR-1.4	High	The system shall identify prescription-only medicines in the cart and shall prevent billing until the associated prescription has been verified.
FR-1.5	Medium	The system shall maintain the Wishlist independently from the Cart. Customers shall be able to move items from the Wishlist to the Cart when required.
FR-1.6	High	The system shall allow new customers to create an account using required details such as name, email address, and password.
FR-1.7	High	The system shall allow registered customers to log in using their registered email address and password.

3.1.2 Billing and Customer Relationship Management

ID	Priority	Requirement
FR-2.1	High	The system shall block checkout for prescription-only medicines until the uploaded prescription has been verified.
FR-2.2	High	If payment fails or is interrupted, the system shall leave the inventory and order state unchanged and shall not perform a partial inventory decrement.
FR-2.3	High	The system shall support UPI, debit card, credit card, and net banking through the configured sandbox payment gateway.
FR-2.4	Medium	The system shall credit MediPoints only after successful confirmation of payment.

FR-2.5	Medium	The system shall route refill reminders to assisted teleconsultation booking rather than automatically redispatching medicines.
FR-2.6	High	The system shall maintain an immutable purchase history for each customer to support reminders and repeat-purchase suggestions.

3.2 Non-Functional Requirements

3.2.1 Security Requirements

ID	Priority	Requirement
NFR-1.1	High	The server shall enforce Role-Based Access Control (RBAC) for every protected request.
NFR-1.2	High	The system shall use bcrypt for password hashing and JWT-based authentication with session expiry.
NFR-1.3	High	The audit log shall be append-only. The system shall not provide update or delete operations for audit records.
NFR-1.4	High	Payment tokens and prescription data shall be encrypted or tokenized at rest.

3.2.2 Performance, Reliability, and Usability Requirements

ID	Priority	Requirement
NFR-2.1	High	Inventory reservation shall be atomic to prevent lost updates and overselling.
NFR-2.2	Medium	The checkout interface shall clearly display the remaining stock-hold time and prescription verification status.
NFR-2.3	Low	The system shall be designed with a 24x7 operational goal. Production uptime guarantees are outside the scope of the academic prototype.

4 Data and Interface Requirements

4.1 Database Entities

The MongoDB database shall contain the following major entities:

- Users
- Medicines / Inventory
- Cart / Wishlist
- Prescriptions
- Orders
- Payments
- AuditLogs
- MediPoints Ledger

4.2 User Interface

The system shall provide:

- A React-based customer interface.
- An Inventory Manager console for pharmacists and inventory managers.
- Cart and Wishlist management.
- Prescription upload and verification status.
- Checkout and payment status.
- Customer purchase history and MediPoints information.

4.3 Application Programming Interface

The backend shall expose RESTful APIs implemented using Node.js and Express.
The APIs shall handle:

- Authentication and authorization.
- Medicine and inventory operations.
- Cart and Wishlist operations.
- Prescription management.
- Order processing.
- Payment processing.
- Customer information and purchase history.
- Audit logging.

4.4 Payment Interface

The system shall integrate with a sandbox/test environment of a payment provider such as Razorpay or Stripe.

No real financial transactions shall be performed by the academic prototype.

4.5 QR/Barcode Interface

QR/barcode functionality shall be simulated in software.

No physical RFID or barcode hardware is required for the prototype.

4.6 Communication Interface

All client-server communication shall use HTTPS.

Authenticated API requests shall use JWT bearer tokens where authentication is required.

4.7 Hardware and Software Requirements (Development & Runtime)

Category	Requirement
Processor	Quad-core processor, 2.5 GHz or higher, recommended for development and local testing.
Memory	8 GB RAM recommended for development and local testing.
Storage	At least 20 GB of available storage for source code, dependencies, build artifacts, logs, and local development tools.
Runtime	Node.js 18 or later; npm or yarn for dependency management.
Database	MongoDB 6 or later, using a local instance or a cloud-hosted tier (e.g., MongoDB Atlas).
Version Control	Git, with per-member commit logging as required by the assignment's documentation policy.
Hosting (Prototype)	The React frontend shall be deployed using Netlify's managed hosting and CDN infrastructure.
Third-Party Accounts	A Razorpay or Stripe developer account in sandbox/test mode.
Client	Any modern browser such as Chrome, Firefox, Edge, or Safari on desktop or mobile.
Transport Security	HTTPS shall be used for all client-server communication.
Secrets Management	JWT signing keys, database connection strings, and payment API keys shall be stored as environment variables and shall never be hardcoded.

5 Future Scope: Regional Demand Pattern Detection

A possible future extension of MediTrack is regional medicine demand pattern detection across participating pharmacies.

The objective would be to identify aggregated demand patterns that could assist pharmacies in inventory planning and supply management.

The following considerations shall apply:

- Regional demand information may be aggregated across participating pharmacies.
- Aggregated information may be used to identify medicine demand patterns.
- The mechanism for sharing such aggregated information remains an open design issue.
- Patient-level information shall not leave the aggregation boundary in non-anonymized form.

5.1 Future Data-Sharing Considerations

Any future regional demand-analysis mechanism shall be designed around anonymization and aggregation so that individual patient information is not exposed.

The detailed data-sharing mechanism is outside the scope of the current prototype.

6 Constraints and Assumptions

6.1 Constraints

ID	Constraint
CON-1	The project must be completed within one academic semester by a three-person development team.
CON-2	Payment processing shall remain restricted to sandbox/test mode. No real transactions shall be performed.
CON-3	QR/RFID functionality shall be simulated. Physical hardware is not required for the prototype.

6.2 Assumptions

ID	Assumption
ASSUME-1	The system assumes that standing refill authorization is not generally available for the intended Indian pharmacy workflow.
ASSUME-2	A stable Internet connection is assumed during checkout.
ASSUME-3	Doctors are assumed to be available for teleconsultation when required by the refill workflow.

6.3 Open Issues

ID	Open Issue
OPEN-1	The mechanism for regional demand aggregation and data sharing has not yet been finalized.

Requirement Traceability

Each requirement shall be mapped to the appropriate module owner during implementation.

Requirement Group	Primary Module	Purpose
FR-1.x	Store / Cart	Medicine catalog, inventory, soft holds, and Wishlist.
FR-2.x	Billing / CRM	Checkout, payment, MediPoints, reminders, and purchase history.
NFR-1.x	Security	Authentication, authorization, data protection, and audit logging.
NFR-2.x	Store / Billing	Atomic inventory handling, checkout usability, and reliability.